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BAGONG PILIPINAS

CC 101 – Introduction to Computing

"From Problem to README": Designing a Number-System Converter and Documenting It in Markdown

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Computer Unit Used (Desktop/Laptop, brand):	**Date:** 10/19/26

I. Objectives

At the end of this activity, you should be able to:

- Analyze a number-system conversion problem by identifying its inputs, processes, loop condition, and outputs.
- Write pseudocode using sequence, selection, and iteration.
- Draw flowcharts using standard symbols and check them by tracing and testing with sample data.
- Create a properly formatted README.md file in Markdown using Notepad, without an Internet connection.
- Upload the README.md to your CC 101 GitHub Computing Portfolio repository (take-home part).

II. Materials Needed

- A computer running Windows 11 with Notepad (no network or Internet connection is needed in the laboratory)
- This worksheet, a pencil, and a ruler (or a flowchart template) for drawing flowcharts
- A USB flash drive, to bring your README.md home for uploading
- For Part G only (outside the laboratory): any device with Internet access and a free GitHub account
- Week 7 lecture notes, for reference on pseudocode, flowchart symbols, and Markdown syntax

III. Background

Programmers do not start by typing code. They first analyze the problem, plan the solution as pseudocode and a flowchart, and check it by hand. In this activity, you will design a program that converts numbers between the decimal and binary number systems you learned in Week 5. An algorithm can be written as pseudocode (English-like steps) or drawn as a flowchart (standard symbols connected by arrows). Good work is also documented. On GitHub, the standard way to introduce a project is a README.md file written in Markdown, a plain-text format that uses symbols such as #, *, and - for headings, emphasis, and lists. Because Markdown is plain text, you can write it in Notepad with no Internet connection and upload it to GitHub later.

IV. Procedure

Part A – Analyze the Problem (Problem 1)

Problem 1: Decimal-to-Binary Converter. Design a program that asks the user for a whole number in **decimal** (base 10) that is 0 or greater, converts it to **binary** (base 2), and displays the binary result.

Worked example (13 in decimal): 13 ÷ 2 = 6 r 1; 6 ÷ 2 = 3 r 0; 3 ÷ 2 = 1 r 1; 1 ÷ 2 = 0 r 1. Reading the remainders from bottom to top gives **1101**.

1. Convert 13 and 25 to binary by hand using the method above, so you understand every step before writing the pseudocode.
2. Complete **Table 1**: list the input, the output, the processes (operations), the decision or loop condition,

and the variables you will need.

Part B – Write the Pseudocode

3. In **Box 1**, write the pseudocode for Problem 1. Use START, INPUT, SET, WHILE … DO … ENDWHILE, IF … THEN … ELSE … ENDIF, DISPLAY, and END, and indent the steps inside the loop.

   Tip: Use **MOD** for the remainder of a division (13 MOD 2 = 1) and **DIV** for whole-number division (13 DIV 2 = 6). Each new remainder must be placed at the **left** of the binary digits found so far, for example SET binary = remainder + binary. Also decide what your algorithm should display when the input is 0.

Part C – Draw the Flowchart and Desk Check

4. In **Figure 1**, draw the flowchart for Problem 1 using the standard symbols: oval (start/end), parallelogram (input/output), rectangle (process), and diamond (decision). Show the loop with an arrow going back to the decision, and label every decision exit **Yes** or **No**.
5. Trace your pseudocode with the input **13** and fill in one row of **Table 2** for each pass through the loop. The last column should end with 1101.
6. Test your pseudocode and flowchart with the five inputs in **Table 3**. First convert each number by hand, then follow your pseudocode or flowchart step by step. If the two results do not match, find and correct the error in your pseudocode and flowchart.

Part D – Reverse Conversion (Problem 2)

Problem 2: Binary-to-Decimal Converter. Design a program that asks the user for a binary number (entered as digits 0 and 1, such as 1101) and displays its decimal value. Use the **positional (place value) method** from Week 5: starting from the rightmost digit, multiply each digit by its place value (1, 2, 4, 8, 16, …) and add the products.

Worked example (1101 in binary): (1 × 1) + (0 × 2) + (1 × 4) + (1 × 8) = 1 + 0 + 4 + 8 = **13**.

7. In **Box 2**, write the pseudocode. You can take the rightmost digit with **binary MOD 10** and remove it with **binary DIV 10**; the place value starts at 1 and doubles after each digit.
8. In **Figure 2**, draw its flowchart.
9. Test your pseudocode with the inputs in **Table 4**, comparing hand computation with the result of following your pseudocode.
10. **Optional / Bonus:** Extend Problem 1 so the user can choose the target base (2, 8, or 16). For base 16, remainders 10 to 15 must be displayed as the letters A to F. Write the extended pseudocode in the Bonus box.

Part E – Create Your README.md Offline

11. Show file extensions so that you can see the real file name. Press **Windows + E** to open File Explorer, then click **View** → **Show** → **File name extensions** (make sure it is checked).
12. In File Explorer, click **Documents**, then click **New** → **Folder**. Name the folder **cc101-portfolio** (all lowercase, with a hyphen) and press **Enter**. Open it and create two more folders inside: **week07-algorithms** and **images**.
13. Press the **Windows key**, type **Notepad**, and press **Enter**. If your Notepad shows a formatting toolbar, open the **View** menu and switch to the Markdown **syntax** view so the symbols you type stay visible.
14. Type the README content below. Replace everything in square brackets with your own information, and replace the pseudocode with your pseudocode from Box 1. Keep the blank lines; they separate the parts.

CC 101 Computing Portfolio

This repository contains my laboratory outputs and documentation for *CC 101 Introduction to Computing* at **Sorsogon State University – Bulan Campus**.

About Me

- **Name:** [Your full name]
- **Program and Section:** BSCS 1-[section]
- **Interests:** [two or three interests]
- **Goal:** [one sentence about your goal as a computer science student]

Course Information

Item	Details
Course	CC 101 Introduction to Computing
Instructor	[Instructor's name]
Academic Year	2026–2027

Portfolio Contents

1. [Week 7 – Pseudocode and Flowcharts](week07-algorithms/)
2. More activities will be added each week.

Sample Pseudocode: Decimal to Binary

```
[Paste your pseudocode from Box 1 here]
```

My Workspace

![My laboratory computer](images/my-pc.jpg)

Last updated: [date in YYYY-MM-DD format]

15. Save the file. Click **File → Save as**. Go to **Documents → cc101-portfolio**. Type **README.md** as the **File name**, set **Save as type** to **All files**, make sure **Encoding** is **UTF-8**, and click **Save**.

16. Verify the file. In File Explorer, open the cc101-portfolio folder and confirm the file is named **README.md**, not README.md.txt. If it ends in .txt, right-click it, choose **Rename**, and remove .txt. Then press the **Windows key**, type **cmd**, press **Enter**, and run the commands below. Record the results in **Table 5**.

    cd %USERPROFILE%\Documents\cc101-portfolio

    dir

    type README.md

    Tip: If “The system cannot find the path specified” appears, your Documents folder may be inside OneDrive. Try **cd %USERPROFILE%\OneDrive\Documents\cc101-portfolio** instead.

17. Optional preview: if your Notepad has a formatted view, switch to it from the **View** menu and note in Table 5 which elements appear formatted. Some elements, such as tables, images, and code blocks, may still appear as plain text. GitHub will display them correctly.

18. Copy the whole **cc101-portfolio** folder to your USB flash drive.

Part F – Peer Review

19. Exchange seats with a classmate. Open their README.md in Notepad and complete the checklist in **Table 6** for their file. Write your name as the reviewer and return the worksheet. Fix any missing element in your own file before leaving.

Part G – Take-Home: Upload to GitHub

Do this part when you have Internet access (at home, on campus Wi-Fi, or on your phone's browser in desktop mode).

20. Go to **github.com** and click **Sign up**. Use an email you check regularly and choose a professional username, such as your name. Verify your email address.

21. Click the **+** button at the upper right and choose **New repository**. Type **cc101-portfolio** as the **Repository name**, choose **Public**, and leave **Add a README file** unchecked. Click **Create repository**.

22. On the setup page, click the link **uploading an existing file**. Drag your README.md (and your week07-algorithms folder, if it has files) onto the page.

23. Under **Commit changes**, type **Add initial README** as the message and click **Commit changes**.

24. Open your repository's main page and check that the README appears formatted. Copy the repository URL (for example, https://github.com/username/cc101-portfolio), record it in **Table 7**, and submit it through the LMS.

    Note: GitHub does not store empty folders. The link to week07-algorithms will work after that folder contains at least one file, and the image will appear after you upload images/my-pc.jpg.

V. Data and Findings

Table 1. Problem Analysis for Problem 1 (Part A)

Item	Your Answer
Restated problem (your own words)	Convert a decimal number into binary using repeated division by 2.
Input	A decimal number, such as 13.
Output	The binary equivalent of the decimal number.
Processes (operations)	Divide by 2, get the remainder, and reverse the remainders.
Loop condition (when the loop stops)	Stop when the quotient becomes 0.
Variables needed	n, remainder, binary
13 and 25 in binary (computed by hand)	13 = 1101; 25 = 11001

Box 1. Pseudocode for Problem 1 (Part B)

```
 START
INPUT n
SET binary = ""
IF n = 0 THEN
   DISPLAY 0
ELSE
   WHILE n > 0 DO
      remainder = n MOD 2
      binary = remainder + binary
      n = n DIV 2
   ENDWHILE
   DISPLAY binary
ENDIF
END
```

Figure 1. Flowchart for Problem 1 (Part C)

Draw your flowchart here.

Table 2. Trace of Problem 1 with Input 13 (Part C)

Pass	n (before)	n MOD 2 (remainder)	n DIV 2 (new n)	binary (so far)

Pass	n (before)	n MOD 2 (remainder)	n DIV 2 (new n)	binary (so far)
1	13	1	6	1
2	6	0	3	01
3	3	1	1	101
4	1	1	0	1101
5	0	—	—	1101

Value displayed at the end: _____

Table 3. Test Cases for Problem 1 (Part C)

Case	Decimal Input	Binary (by hand)	Binary From Your Pseudocode	Match? (Y/N)
1	0	0	0	Y
2	5	101	101	Y
3	13	1101	1101	Y
4	25	11001	11001	Y
5	255	11111111	11111111	Y

Box 2. Pseudocode for Problem 2 (Part D)

```
 START
INPUT binary
SET decimal = 0
SET power = 0
WHILE binary > 0 DO
    digit = binary MOD 10
    decimal = decimal + digit * (2 ^ power)
    binary = binary DIV 10
    power = power + 1
ENDWHILE
DISPLAY decimal
END
```

Figure 2. Flowchart for Problem 2 (Part D)

Draw your flowchart here.

Table 4. Test Cases for Problem 2 (Part D)

Case	Binary Input	Decimal (by hand)	Decimal From Your Pseudocode	Match? (Y/N)
1	1101	13	13	Y
2	101010	42	42	Y
3	11111111	255	255	Y
4	10000000	128	128	Y

Bonus Box. Extended Pseudocode: Decimal to Base 2, 8, or 16 (Optional)

```
 START
INPUT n
INPUT base
SET digits = "0123456789ABCDEF"
SET result = ""
IF n = 0 THEN
   DISPLAY 0
ELSE
   WHILE n > 0 DO
      remainder = n MOD base
      result = digits[remainder] + result
      n = n DIV base
   ENDWHILE
   DISPLAY result
ENDIF
END
```

Table 5. README.md File Check (Part E)

Item	**Your Findings**
Full folder path of cc101-portfolio	C:\Users\[YourUsername]\Documents\cc101-portfolio
File name shown in File Explorer	README.md
Type shown in File Explorer (Details view)	Markdown File
Size of README.md shown by dir (bytes)	Depends on your file
Does type README.md show your Markdown text? (Y/N)	Y
Notepad formatted view available? (Y/N)	Y
Elements that appeared formatted in Notepad (if available)	Headings, bold text, and bullet points may appear as plain Markdown symbols.

Table 6. Peer Review Checklist (Part F)

Reviewer's name: _____ Owner of the README: _____

Element	**Markdown Used**	**Present and Correct? (Y/N)**
Level-1 heading	#	Y
Level-2 headings	##	N
Paragraph	Text with a blank line before and after	Y

Element	Markdown Used	Present and Correct? (Y/N)
Bold and italic text	****bold** *italic***	Y
Bulleted list	**- item**	Y
Numbered list	**1. item**	N
Table	| **---** | header separator	Y
Link	**[text](url)**	Y
Image	**![alt](path)**	Y
Code block	Lines between two lines of three backticks	Y
Horizontal rule	**---**	Y
File saved as README.md (not .txt)	_	N

Table 7. GitHub Submission (Part G, Take-Home)

Item	Your Answer
GitHub username	My GiitHub username
Repository URL	My GitHub repository link
Date uploaded (YYYY-MM-DD)	Date when you uploaded files
Does the README display formatted on GitHub? (Y/N)	Y, if correctly uploaded
Elements that did not display as expected, and why	None, if everything displays correctly.

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Which was easier for you to create, the pseudocode or the flowchart? Why?

It was easy to understand the process before drawing the flowchart.

2. Did tracing and testing reveal any errors in your pseudocode or flowcharts? Describe what you found

and how you corrected it.

Yes, testing helped me find mistakes in the loop and calculations. I corrected the steps and checked the results again to make sure the binary conversion is correct.

3. How is writing a README in Markdown different from typing a document in Microsoft Word?

Microsoft Word uses formatting tools and buttons to change the appearance of a document.

4. How can a GitHub portfolio help you as a future computing professional?

A GitHub portfolio can show my programming activities, projects, and skills. It can help me improve my coding abilities

VII. Scoring Guide

Criteria	Description	Points
Problem Analysis & Pseudocode	Table 1 is complete; the pseudocode for both conversions is correct (correct use of MOD, DIV, and the loop), uses proper keywords, and is indented.	25
Flowcharts & Testing	Both flowcharts use the correct symbols with labeled decisions and a clear loop; the trace (Table 2) and test cases (Tables 3 and 4) are correct.	25
README.md	The file is saved correctly as README.md, uses all required Markdown elements, and passes the peer review in Table 6.	25
GitHub Upload	The README is uploaded to a public cc101-portfolio repository and the URL is submitted (Table 7).	10
Realization / Reflection	Answers are thoughtful, specific to the student's own work, and well written.	10
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	5
Total		**100**